F. Additional experiments in SEDD (90M) vs. GPT-2 (86M)

F.1. Ablation studies

In this section, we first provide additional ablation studies for SEDD (90M) vs GPT-2 (86M) with different training and dataset settings (Fig 21 and Fig 22).

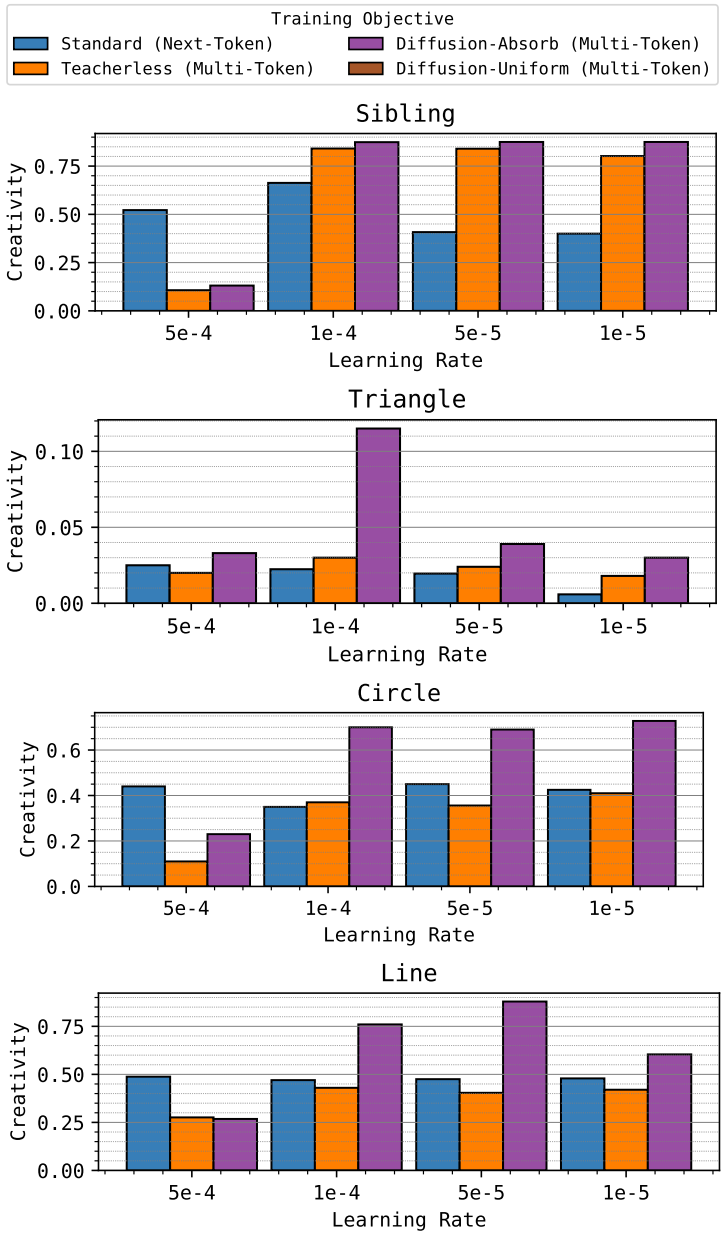

Figure 21. **Learning rates and algorithmic creativity for the SEDD (90M) model vs. GPT-2 (86M):** MTP achieves higher algorithmic creativity than NTP when both are trained at their optimal learning rates. We use greedy decoding in this plot.

F.2. Effect of seed length

We provide an ablation study on the seed length for NTP vs MTP on the Sibling Discovery task (Fig 23 (left)). We see that longer seeds lead to higher algorithmic creativity.

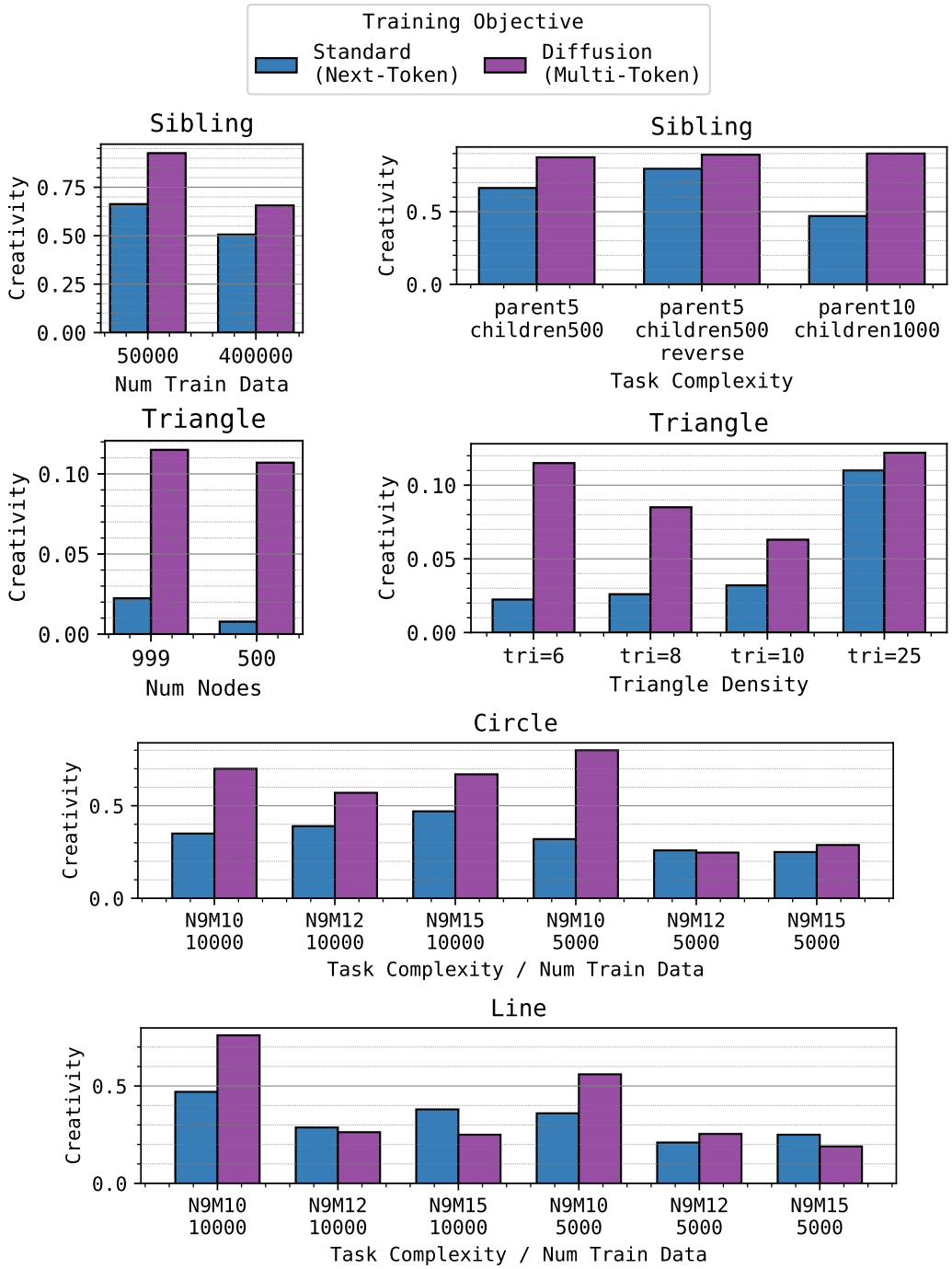

Figure 22. **Task complexity and algorithmic creativity of `SEDD (90M)` model vs. `GPT-2 (86M)`:** `MTP` consistently outperforms `NTP` under varying task configurations, with some exceptions in the `Line Construction`and `Circle Construction`datasets. We use greedy decoding in this plot.

F.3. Seed-conditioning for diffusion training

In Fig 23 (right), we apply seed-conditioning to the `SEDD (90M)` on the `Sibling Discovery` task. Unlike what we see on transformers, we do not observe improved creativity when applying seed-conditioning to diffusion training.

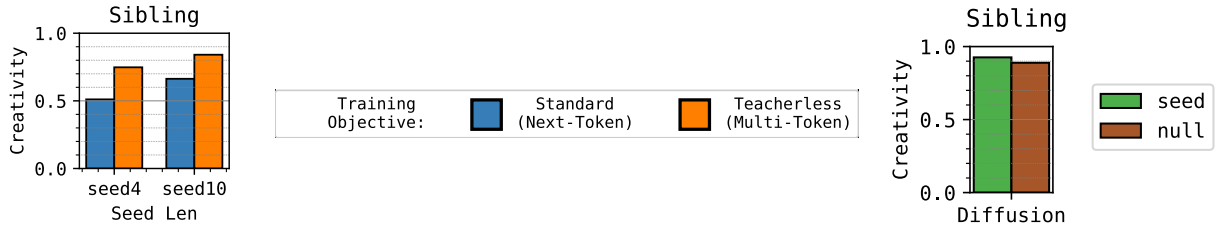

Figure 23. `GPT-2 (86M)` **Transformer achieves higher algorithmic creativity with longer seeds (top):** We report algorithmic creativity with seeds of length 4 and 10, with both `NTP` and teacherless `MTP` (with greedy decoding). **We do not see gains of seed-conditioning when it comes to diffusion training (right):** We report algorithmic creativity of the `SEDD (90M)` model with or without seed-conditioning.

F.4. Effect of top-K sampling

Figure 24 reports the algorithmic creativity scores of `NTP` vs teacherless `MTP` training with seed-conditioning and top-K sampling where K=50.

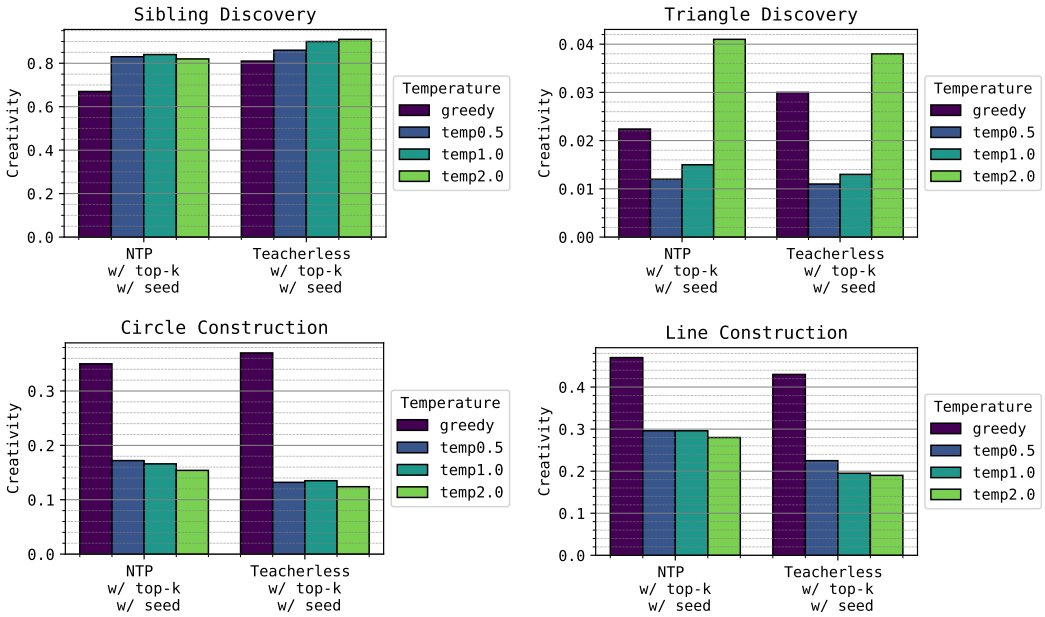

Figure 24. **Creativity scores for `NTP` vs teacherless `MTP` training with seed-conditioning + top-K sampling where $K = 50$.**

F.5. Format sensitivity for `Triangle Discovery`

Recall that our input format for `Triangle Discovery` follows the *edge list representation* of triangles (§C, Fig. 10). For instance, triangle ABC is represented as `AB, BC, CA`. This format explicitly lists the edges of the triangle, making it easier for the model to attend to edge-level patterns during learning.

We also experimented with an alternative *node-based representation*, where triangle ABC is represented more compactly as `ABC`, without making the edges explicit. We note in Fig 26 that the models are curiously sensitive to the way the triangles are formatted. Models trained on the node-based format perform equally badly with all training objectives, while the diffusion model outperforms `NTP` by a large margin with the edge list representation.

F.6. Additional ablations

Figure 28 reports the ablations on `GPT-2 (86M)` for (1) `NTP` vs `MTP`, (2) effect of seed conditioning, (3) effect of top-K, and (4) effect of temperature sampling.